



Human Computer Interaction

COMP322

Section:1

“Phase3: Evaluation and revision”

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1. Usability Test Protocol

1.1 Objective

The objective of this usability test will be to evaluate the new redesign of the QOU Student Portal and analyze whether the new design enhances usability, reduces the cognitive load, and allows the students to perform their academic activities effectively. The test will revolve around the redesign objectives that were identified during phase one of this project.

1.2 Participants

Three students from the university who were not included in the Phase 1 interviews were selected for the usability testing exercise. The participants had used the university portals before and came from various disciplines.

Participant	Age	Major	Experience
P1	20	Arabic literature	Frequent User
P2	22	Business Administration	Moderate User
P3	19	accounting	Frequent User

1.3 Testing Environment

- Device: Laptop/Desktop.
- Prototype: Interactive Figma Prototype of the redesigned QOU Student Portal.
- Method: Think-Aloud Protocol.
- Duration: Approximately 15 minutes per participant.

Note: Participants were asked to speak their thoughts, expectations, and difficulties loudly while completing the assigned tasks.

1.4 Testing Procedure

1. The participants were briefed about the aim of the experiment and signed the consent form.
2. The participants made use of the interactive prototype independently without help or directions.
3. The participants performed a series of pre-defined tasks while following the think aloud method.
4. The evaluators observed and noted down the task performance, participants thoughts and expectations, time taken to complete the task, the number of errors occurred, and feedback of the participants after the process.
5. Towards the conclusion of the experiment, the participants filled up the SUS form.
6. The quantitative data along with the qualitative data were finalized and analyzed.

1.5 Test Tasks

Participants were asked to complete the following seven tasks using the redesigned prototype while thinking aloud.

Task 1: Check Up Today's Information:

Scenario: You have just logged into the portal and want to know what is important today.

Task: Find today's classes and check whether there is any new message or published grades.

Success Criteria: The participant successfully identifies today's classes and notices any new updates.

Task 2: View Next Week's Schedule

Scenario: You want to know your classes for next week.

Task: Open the Semester Schedule page and display the schedule for next week.

Success Criteria: The participant successfully sees the schedule for next week.

Task 3: See Course Information Details

Scenario: You need to see a course's information.

Task: Open the Semester Schedule page and view the details of one of the courses.

Success Criteria: The participant successfully opens the course details pop-up and identifies the course information.

Task 4: Check Course Grades

Scenario: You want to check your grades in a course.

Task: Open Course Services, and find your Overall Grade and Class Average.

Success Criteria: The participant successfully identifies both the overall grade and class average.

Task 5: Check Assignment Status

Scenario: You want to know whether you still have pending assignments.

Task: Open the Assignments tab and identify which assignments are still pending.

Success Criteria: The participant correctly identifies all assignments that have not been submitted.

Task 6: Access Learning Materials

Scenario: You need to access the learning materials for one of your courses.

Task: Open Course Services, find and access the Course Material link.

Success Criteria: The participant successfully reaches the Course Material link.

Task 7: Check Upcoming Exams

Scenario: You want to know when and where your next exam will be.

Task: Open Exams schedule, and find the date, time, and location of the next exam.

Success Criteria: The participant successfully identifies the next exam and correctly reports its date, time, and location.

2. Quantitative Results

To evaluate the usability of the redesigned QOU Student Portal, quantitative measures including task completion rate, time on task, and error count were collected for each participant.

2.1 Time on Task (seconds)

Task	P1	P2	P3	Average
Task 1	18	20	17	18.3
Task 2	24	26	22	24.0
Task 3	21	24	20	21.7
Task 4	32	35	29	32.0
Task 5	27	30	26	27.7
Task 6	19	22	18	19.7
Task 7	23	25	21	23.0

2.2 Task Completion Rate & Error Count

Task	Completion Rate	Total Errors
Task 1	100%	0
Task 2	100%	1
Task 3	100%	0
Task 4	100%	2
Task 5	100%	1
Task 6	100%	0
Task 7	100%	1

All participants successfully completed all seven tasks, achieving a 100% task completion rate. Average task completion time ranged from 18.3 to 32.0 seconds, with the grades task taking longest due to the change in reducing the number of cards to only one card with a dropdown menu to select the course. The low error counts across all tasks suggest that the redesigned interface was intuitive and easy to navigate.

3. Qualitative Findings

Several consistent themes were found from the think-aloud sessions across all three participants.

Theme 1: Easier Access to Information

All participants appreciated having important academic information available directly on the dashboard. They were able to identify today's classes and recent updates without navigating through multiple screens.

P1: "Everything I need is visible immediately".

P3: "I do not need to keep moving from one page to another anymore?"

Theme 2: Improved Schedule Navigation

Participants found the redesigned schedule significantly easier to use because the portal automatically displayed the current week's sessions, eliminating the need to calculate whether the week was odd or even.

P2: "This is much better than calculating odd and even weeks mentally".

Theme 3: Clear Course Services Organisation

Participants moved easily between the Exams, Assignments, Grades, and Links tabs within Course Services. The tab structure reduced navigation effort and made related information intuitively grouped.

P1: "The tabs are so simple and everything is organized".

Theme 4: Better Understanding of Grades and Assessments

Participants appreciated the clear presentation of individual grades and class averages. The dynamic assessment names per course helped them identify exactly which item they were reviewing.

P3: "I can understand my performance immediately without opening multiple pages".

Overall, participants reported that the redesigned portal was simple to use, required fewer navigation steps, and significantly reduced the effort needed to access important academic information.

4. System Usability Scale (SUS) Results

At the end of each usability testing session, participants completed the System Usability Scale (SUS) questionnaire to evaluate their overall experience with the redesigned QOU Student Portal.

Participant	SUS Score
P1	85
P2	82.5
P3	87.5

Average SUS Score = $(85 + 82.5 + 87.5) \div 3 = 85$

The SUS scores for the newly designed QOU student portal averaged 85 points out of 100. Based on the SUS scoring guide: a score from 80 to 90 points is deemed Excellent and suggests very high user satisfaction. These results show that the users found the newly designed portal to be very easy to learn and navigate. In addition, it performed efficiently in accomplishing common academic activities. It further surpassed the minimum acceptable score of 70 points as set in Phase 1 usability requirements R7.

5. Heuristic Re-evaluation

A heuristic re-evaluation was conducted to determine whether the redesigned QOU Student Portal successfully addressed the usability violations identified during Phase 1. The prototype was revised by using Nielsen's Ten Usability Heuristics.

Phase 1 Problem	Heuristic(s)	Redesign Improvement	Result
Dense schedule tables with unnecessary columns	H8	Irrelevant columns removed; extra details moved to course details pop-ups.	Resolved
Students had to calculate odd/even weeks manually	H2, H6, H7, H8	The system automatically displays the current week; week selector added for future weeks.	Resolved
Course services repeated for every course	H8	Single Course Services section with course dropdown introduced.	Resolved
Meeting status and online sessions were unclear	H1, H10	Clear status messages added for unscheduled or online meetings.	Improved
Students had to open every exam/assignment panel separately	H1, H5, H7	Status indicators highlight new exams and assignments from the dashboard.	Resolved
Exam schedule required filtering before showing results	H7, H8	Upcoming exams shown immediately; filtering is now optional.	Resolved
Exam review showed error messages with no status indication	H5, H7	Review availability clearly indicated via enabled/disabled button states.	Resolved
Students had to open each course to check for published grades	H1, H5, H7	Grade publication status visible via dashboard indicators and three-state labels.	Resolved

Phase 1 Problem	Heuristic(s)	Redesign Improvement	Result
Assessment names were static and generic	H2, H10	Assessment names are now dynamic and course-specific.	Resolved

The re-evaluation confirms that 8 out of 9 identified violations were fully resolved and one was meaningfully improved. No new heuristic violations were introduced by the redesign.

6. Revision Plan

Although the usability evaluation showed that the redesigned **QOU Student Portal** achieved high usability and strongly positive user feedback, several improvements could be prioritised in future iterations, listed below in order of estimated impact.

1. Students get Emails when exams are announced or course grades get published (Highest Priority):

This was the single most requested feature across all Phase 1 interviews. Students log in multiple times per day during exam periods just to check whether grades have been posted. Adding push notifications for grade publication, new assignments, and exam schedule changes would directly eliminate this frustrating daily behaviour.

2. Dashboard Customisation (Medium Priority):

Different students prioritise different information. Allowing users to reorder or pin dashboard cards would personalise the experience without adding cognitive complexity for those who prefer the default layout.

3. Contextual Help and Onboarding for New Students (Medium Priority):

Expandable tooltips explaining assessment types, and portal features would reduce the support needed from other students or teachers and improve first-semester student confidence.

4. Advanced Filtering and Search Across Semesters (Lowest Priority):

Students might need to look up grades or exams from previous semesters. Adding a cross-semester search bar in the Grades and Schedule sections would reduce the need for office visits to retrieve archived academic records.

These revisions would complete the UCD loop by acting on the data collected in this phase rather than treating the prototype as a finished product.

7. Reflection

This section is about what we learned through the full User-Centred Design process across the three phases, and how experience differed from our expectations.

7.1 What We Learned from the UCD Process

Before starting this project, we thought that UCD is a methodological checklist that goes: conduct interviews, build personas and scenarios, design, test. What we did not understand was how each phase would effect the next one. The heuristic evaluation in **Phase 1** identified many violations, but the interviews revealed which ones actually bothered the students the most. **Violation 3** (the odd/even week system, rated 4/4) was the most severe on paper, and the interviews confirmed it. Every participant had a story about being confused or missing a class because of it. That alignment between our evaluation and user experience gave us confidence that the problem was real and severe, not just a target for a university student trying to find issues for his project.

Phase 2 forced us to make design decisions rather than simply list problems. Translating interview findings and design solutions from phase one into well organized and design decision self-explanatory wireframes. Especially the dashboard summary card that summarized our work along with the three-state status indicators and the odd/even week solution. We learned that good design is not about adding features but about removing the burden and simplifying the original system.

7.2 What Was Harder Than Expected

The most difficult part of the project was the usability testing phase. We expected participants to encounter the same problems we had identified. Instead, we found that some issues we considered obvious were immediately understood by participants, while Task 4 (checking grades) produced the most errors and the longest average time, not because of a design flaw, but because the detailed grade breakdown contained more information than participants expected to see. This taught us that adding information is not always neutral "even well-designed content takes time to process".

Recruiting participants who had not been involved in Phase 1 and scheduling sessions within the 15-minute window while allowing participants to think aloud freely required more facilitation skill than we initially estimated, especially that we didn't want to direct the participant or to provide extra or less information that required to them.

7.3 How Real User Research Differed from Our Initial Assumptions

At the start of the project, we assumed that students' main frustration with the QOU portal was its visual design, especially that it looked outdated and cluttered(to be honest we made fun of it). The interviews quickly corrected this assumption. Participants rarely mentioned appearance. What frustrated them were redundant processes such as not knowing whether a grade had been published or if an exam or assignment got posted, missing a class because the odd/even week type was unclear and needing memorization, and having to visit the registration office in person for information that should have been available online.

We also assumed that participants from different academic programmes would have significantly different needs. In practice the core frustrations were remarkably consistent across

students from different fields and that the actual differences in needs was due to the personas we made, not to stereotypes or hypotheses. This gave us confidence that designing for the primary persona (Mohammad) while considering and trying to satisfy the secondary persona would genuinely serve the broader student population.

7.4 Conclusion

The Phase 3 evaluation validated the redesign's core decisions. A 100% task completion rate alongside with an average SUS score of 85/100 which is much higher than the 70 point threshold set in Phase 1 requirements. The previous two indicators confirm that the new portal design improves usability, reduces cognitive load, improves access to academic information and eliminates the need for redundant checks and navigation through the whole portal. Also, this process taught us that an effective UCD process begins with the research phase not with the design phase because the quality of the redesign was directly led by the quality of our understanding of the students needs, use cases and backgrounds instead of hypotheses that the designer or the developer might make.

person 1

Appendix D: System Usability Scale (SUS) questionnaire

The SUS is a standard 10-item questionnaire used in usability testing, developed by John Brooke (1986). It produces a single score between 0 and 100 representing overall perceived usability. Use this in Phase 3 after each participant completes the usability test.

Participants respond on a 5-point scale from Strongly Disagree (1) to Strongly Agree (5). For each statement, circle the number that best reflects your view.

#	Statement	SD	D	N	A	SA
1	I think that I would like to use this system frequently.	1	2	3	4	5
2	I found the system unnecessarily complex.	1	2	3	4	5
3	I thought the system was easy to use.	1	2	3	4	5
4	I think that I would need the support of a technical person to be able to use this system.	1	2	3	4	5
5	I found the various functions in this system were well integrated.	1	2	3	4	5
6	I thought there was too much inconsistency in this system.	1	2	3	4	5
7	I would imagine that most people would learn to use this system very quickly.	1	2	3	4	5
8	I found the system very cumbersome to use.	1	2	3	4	5
9	I felt very confident using the system.	1	2	3	4	5
10	I needed to learn a lot of things before I could get going with this system.	1	2	3	4	5

Scoring instructions (for evaluators)

1. For odd-numbered items (1, 3, 5, 7, 9): subtract 1 from the participant's score.
2. For even-numbered items (2, 4, 6, 8, 10): subtract the participant's score from 5.
3. Sum the ten adjusted scores and multiply by 2.5.
4. The result is a score between 0 and 100.
5. Average across all participants for the final SUS score.

Interpretation

Based on Bangor, Kortum and Miller (2008):

Score range	Grade
Below 50	Poor usability
50-70	Acceptable but needs improvement
70-80	Good
80-90	Excellent
Above 90	Best imaginable

person 2

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Appendix C: Participant consent form

A consent form must be signed by every interview and usability test participant before the session begins. This template may be used directly or adapted. Teams are encouraged to translate it into Arabic for Arabic-speaking participants; both versions can be offered.

COMP322: Human Computer Interaction

Participant Consent Form

Project title: Q U student portal:-
Team members: Eman, donia, Osama, Salo.

Course instructor: Dr Hisham Ihshaish, Department of Computer Science, Birzeit University

What is this study about?

We are students in a Human-Computer Interaction course at Birzeit University. As part of our course project, we are studying how people use [insert system or service name] and looking for ways to improve it.

What will you be asked to do?

You will be asked to (choose one):

- Take part in an interview of approximately 20-45 minutes about your experience with the system.
- Use a prototype of a redesigned system while we observe, for approximately 30 minutes, and answer a short questionnaire afterwards.

We may take written notes, and with your permission, record audio or video. Recordings will only be used by our team for analysis and will be deleted at the end of the course.

What are the risks and benefits?

There are no known risks beyond those of a normal conversation or software use. There is no direct benefit to you, but your input will help us design a better version of the system.

How will your information be used?

- Your identity will not appear in any report or presentation. We will refer to you by a code such as "Participant 1".
- Direct quotes may be used, but only in ways that cannot identify you.
- Data will be stored securely and accessible only to the team and the instructor.
- All raw data will be deleted at the end of the academic semester.

Your rights

- Participation is entirely voluntary.
- You may decline to answer any question.

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COMP322: Project and Assignments Specification

- You may withdraw at any time without giving a reason and without any consequence. Any data already collected from you will be deleted.

Consent

I have read and understood the information above. I agree to take part in this study.

- ☒ I agree to participate.
- ☐ I agree to be audio or video recorded.
- ☐ I agree to have direct quotes used anonymously in the team's report and presentation.

Participant name: Salma
Signature: [Signature]
Date: 2026
Team member witness: Salma

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Participant name: Sahar
Signature: [Signature]
Date: 21-6
Team member witness: [Signature]

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Participant name: Salwa Al-Sli
Signature: [Signature]
Date: 19-6
Team member witness: Donia Al-Sli